

Week 2 Assignment

NumPy & Pandas for Data Analytics

Objective

This report demonstrates practical use of NumPy and Pandas for arrays, numerical computation, DataFrames, inspection, filtering, missing-value handling, dataset combination, GroupBy, Pivot Tables, exporting, and a mini data analysis project.

Tools: Python, NumPy, Pandas, Matplotlib, ReportLab

Datasets: Student Performance, Employees, Sales, and Missing Values CSV files included in the submission package.

Task 1 — Created NumPy arrays, displayed shape, size and data type, and demonstrated 1-D and 2-D arrays.

Source Code

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90, 100])
print("NumPy Array:", arr)
print("Shape:", arr.shape)
print("Size:", arr.size)
print("Data Type:", arr.dtype)

one_d = np.array([1, 2, 3, 4, 5])
two_d = np.array([[1, 2, 3], [4, 5, 6]])

print("\nOne-Dimensional Array:", one_d)
print("Two-Dimensional Array:\n", two_d)
```

Program Output

```
NumPy Array: [ 10  20  30  40  50  60  70  80  90 100]
Shape: (10,)
Size: 10
Data Type: int64

One-Dimensional Array: [1 2 3 4 5]
Two-Dimensional Array:
[[1 2 3]
 [4 5 6]]

[stderr]
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/patches/warm_spreadsheet_runtime_on_start
rtup.py", line 26, in warm_spreadsheet_runtime_on_startup
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 785, in wa
rm_spreadsheet_runtime
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 720, in _w
arm_feature_flows
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 704, in _w
arm_collaboration_flows
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/generated/interface/models.py", line 32
317, in hydrate_crdt_from_proto
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/remote.py", line 749, in __call__
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/client.py", line 150, in call
artifact_tool.rpc.client.RemoteError: hydrateCrdtFromProto requires an empty collaborative document.
```

Task 2 — Demonstrated indexing, slicing, row/column access and reshaping between multiple dimensions.

Source Code

```
import numpy as np

arr = np.arange(1, 13)
print("Original 1-D Array:", arr)

print("Element at index 3:", arr[3])
print("Element at index -1:", arr[-1])
print("Slice arr[2:7]:", arr[2:7])

matrix = arr.reshape(3, 4)
print("\nOriginal 2-D Array:\n", matrix)
print("First row:", matrix[0])
print("Second column:", matrix[:, 1])
print("Element at row 2, column 3:", matrix[1, 2])

reshaped_2x6 = arr.reshape(2, 6)
reshaped_4x3 = arr.reshape(4, 3)

print("\nReshaped to 2 x 6:\n", reshaped_2x6)
print("\nReshaped to 4 x 3:\n", reshaped_4x3)
```

Program Output

```
Original 1-D Array: [ 1  2  3  4  5  6  7  8  9 10 11 12]
Element at index 3: 4
Element at index -1: 12
Slice arr[2:7]: [3 4 5 6 7]
```

```
Original 2-D Array:
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
First row: [1 2 3 4]
Second column: [ 2  6 10]
Element at row 2, column 3: 7
```

```
Reshaped to 2 x 6:
[[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]]
```

```
Reshaped to 4 x 3:
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]]
```

```
[stderr]
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/patches/warm_spreadsheet_runtime_on_start
rtup.py", line 26, in warm_spreadsheet_runtime_on_startup
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 785, in wa
rm_spreadsheet_runtime
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 720, in _w
arm_feature_flows
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 704, in _w
arm_collaboration_flows
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/generated/interface/models.py", line 32
317, in hydrate_crdt_from_proto
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/remote.py", line 749, in __call__
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/client.py", line 150, in call
artifact_tool.rpc.client.RemoteError: hydrateCrdtFromProto requires an empty collaborative document.
```

Task 3 — Performed addition, subtraction, multiplication, division, mean, median, minimum, maximum, standard deviation and sum.

Source Code

```
import numpy as np

data = np.array([12, 18, 25, 30, 22, 35, 40, 28, 15, 32], dtype=float)
other = np.array([2, 3, 5, 6, 2, 5, 4, 7, 3, 4], dtype=float)

print("Dataset:", data)
print("\nMathematical Operations")
print("Addition:", data + other)
print("Subtraction:", data - other)
print("Multiplication:", data * other)
print("Division:", np.round(data / other, 2))

print("\nStatistical Operations")
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Minimum:", np.min(data))
print("Maximum:", np.max(data))
print("Standard Deviation:", round(np.std(data), 2))
print("Sum:", np.sum(data))
```

Program Output

```
Dataset: [12. 18. 25. 30. 22. 35. 40. 28. 15. 32.]
Mathematical Operations
Addition: [14. 21. 30. 36. 24. 40. 44. 35. 18. 36.]
Subtraction: [10. 15. 20. 24. 20. 30. 36. 21. 12. 28.]
Multiplication: [ 24. 54. 125. 180. 44. 175. 160. 196. 45. 128.]
Division: [ 6.  6.  5.  5. 11.  7. 10.  4.  5.  8.]

Statistical Operations
Mean: 25.7
Median: 26.5
Minimum: 12.0
Maximum: 40.0
Standard Deviation: 8.54
Sum: 257.0

[stderr]
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/patches/warm_spreadsheet_runtime_on_start
rtup.py", line 26, in warm_spreadsheet_runtime_on_startup
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 785, in wa
rm_spreadsheet_runtime
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 720, in _w
arm_feature_flows
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 704, in _w
arm_collaboration_flows
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/generated/interface/models.py", line 32
317, in hydrate_crdt_from_proto
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/remote.py", line 749, in _call_
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/client.py", line 150, in call
    artifact_tool.rpc.client.RemoteError: hydrateCrdtFromProto requires an empty collaborative document.
```

Task 4 — Created a Pandas Series and DataFrame, inspected columns/index and added a new column.

Source Code

```
import pandas as pd

series = pd.Series([85, 91, 78, 88, 95], index=["A", "B", "C", "D", "E"])
print("Pandas Series:\n", series)

df = pd.DataFrame({
    "Name": ["Aarav", "Diya", "Rohan", "Ananya", "Kabir"],
    "Department": ["CSE", "CSE", "AIIML", "AIIML", "CSE"],
    "Marks": [88, 76, 91, 67, 84]
})

print("\nOriginal DataFrame:\n", df)
print("\nColumn Names:", list(df.columns))
print("Index:", df.index)

df["Result"] = ["Pass", "Pass", "Pass", "Pass", "Pass"]
print("\nUpdated DataFrame:\n", df)
```

Program Output

Pandas Series:

A 85
B 91
C 78
D 88
E 95
dtype: int64

Original DataFrame:

	Name	Department	Marks
0	Aarav	CSE	88
1	Diya	CSE	76
2	Rohan	AIIML	91
3	Ananya	AIIML	67
4	Kabir	CSE	84

Column Names: ['Name', 'Department', 'Marks']
Index: RangeIndex(start=0, stop=5, step=1)

Updated DataFrame:

	Name	Department	Marks	Result
0	Aarav	CSE	88	Pass
1	Diya	CSE	76	Pass
2	Rohan	AIIML	91	Pass
3	Ananya	AIIML	67	Pass
4	Kabir	CSE	84	Pass

[stderr]

Spreadsheet runtime warmup failed during python startup

Traceback (most recent call last):

File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/patches/warm_spreadsheet_runtime_on_start
rtup.py", line 26, in warm_spreadsheet_runtime_on_startup
File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 785, in wa
rm_spreadsheet_runtime
File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 720, in _w
arm_feature_flows
File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 704, in _w
arm_collaboration_flows
File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/generated/interface/models.py", line 32
317, in hydrate_crdt_from_proto
File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/rpc/remote.py", line 749, in __call__
File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/rpc/client.py", line 150, in call
artifact_tool.rpc.client.RemoteError: hydrateCrdtFromProto requires an empty collaborative document.

Task 5 — Read a CSV and used head, tail, shape, columns, dtypes, info and describe.

Source Code

```
import pandas as pd

df = pd.read_csv("../datasets/student_performance.csv")

print("First 5 Rows:\n", df.head())
print("\nLast 5 Rows:\n", df.tail())
print("\nNumber of Rows and Columns:", df.shape)
print("\nColumn Names:", list(df.columns))
print("\nData Types:\n", df.dtypes)
print("\nINFO")
df.info()
print("\nDESCRIPTIVE STATISTICS\n", df.describe())
```

Program Output

```
First 5 Rows:
# Student ID      Name Department      Marks      Attendance      City
0         101      Aarav      CSE         88         92      Kanpur
1         102      Diya      CSE         76         85      Lucknow
2         103      Rohan      AIMA         91         96      Kanpur
3         104      Ananya      AIMA         87         78      Delhi
4         105      Kabir      CSE         84         90      Kanpur

Last 5 Rows:
# Student ID      Name Department      Marks      Attendance      City
5         106      Ishita      ECE         72         82      Lucknow
6         107      Arjun      AIMA         95         98      Delhi
7         108      Mesra      CSE         81         88      Kanpur
8         109      Vivaan      ECE         69         75      Agra
9         110      Sana      AIMA         89         94      Delhi

Number of Rows and Columns: (10, 6)
Column Names: ['Student_ID', 'Name', 'Department', 'Marks', 'Attendance', 'City']

Data Types:
Student_ID      int64
Name            object
Department      object
Marks           int64
Attendance      int64
City           object
dtype: object

INFO
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
--  --
0   Student_ID  10 non-null      int64
1   Name        10 non-null      object
2   Department  10 non-null      object
3   Marks       10 non-null      int64
4   Attendance  10 non-null      int64
5   City        10 non-null      object
dtypes: int64(3), object(3)
memory usage: 612.0+ bytes

DESCRIPTIVE STATISTICS
# Student ID      Marks      Attendance
count      10.000000      10.000000      10.000000
mean       105.500000      81.200000      87.800000
std         3.627651         9.735631         7.699328
min        101.000000      67.000000      75.000000
25%        103.250000      73.000000      82.750000
50%        105.500000      82.500000      89.000000
75%        107.750000      88.750000      93.500000
max         110.000000     95.000000     98.000000
```

Task 6 — Selected columns/rows, applied single and multiple filters, and sorted in ascending and descending order.

Source Code

```
import pandas as pd

df = pd.read_csv("../datasets/student_performance.csv")

print("Selected Columns:\n", df[["Name", "Department", "Marks"]])
print("\nSelected Rows (first 3):\n", df.iloc[:3])

print("\nStudents with Marks >= 85:\n", df[df["Marks"] >= 85])

filtered = df[(df["Marks"] >= 80) & (df["Attendance"] >= 90)]
print("\nStudents with Marks >= 80 AND Attendance >= 90:\n", filtered)

print("\nSorted by Marks - Ascending:\n", df.sort_values("Marks"))
print("\nSorted by Marks - Descending:\n", df.sort_values("Marks", ascending=False))
```

Program Output

Selected Columns:

	Name	Department	Marks
0	Aarav	CSE	88
1	Diya	CSE	76
2	Rohan	AIML	91
3	Ananya	AIML	67
4	Kabir	CSE	84
5	Ishita	ECE	72
6	Arjun	AIML	95
7	Meera	CSE	81
8	Vivaaan	ECE	69
9	Sana	AIML	89

Selected Rows (first 3):

Student ID	Name	Department	Marks	Attendance	City
0	101	Aarav	CSE	88	Kanpur
1	102	Diya	CSE	76	Lucknow
2	103	Rohan	AIML	91	Kanpur

Students with Marks >= 85:

Student ID	Name	Department	Marks	Attendance	City
0	101	Aarav	CSE	88	Kanpur
2	103	Rohan	AIML	91	Kanpur
6	107	Arjun	AIML	95	Delhi
9	110	Sana	AIML	89	Delhi

Students with Marks >= 80 AND Attendance >= 90:

Student ID	Name	Department	Marks	Attendance	City
0	101	Aarav	CSE	88	Kanpur
2	103	Rohan	AIML	91	Kanpur
4	105	Kabir	CSE	84	Kanpur
6	107	Arjun	AIML	95	Delhi
9	110	Sana	AIML	89	Delhi

Sorted by Marks - Ascending:

Student ID	Name	Department	Marks	Attendance	City
3	104	Ananya	AIML	67	Delhi
8	109	Vivaaan	ECE	69	Agra
5	106	Ishita	ECE	72	Lucknow
1	102	Diya	CSE	76	Lucknow
7	108	Meera	CSE	81	Kanpur
4	105	Kabir	CSE	84	Kanpur
0	101	Aarav	CSE	88	Kanpur
9	110	Sana	AIML	89	Delhi
2	103	Rohan	AIML	91	Kanpur
6	107	Arjun	AIML	95	Delhi

Sorted by Marks - Descending:

Student ID	Name	Department	Marks	Attendance	City
6	107	Arjun	AIML	95	Delhi
2	103	Rohan	AIML	91	Kanpur
9	110	Sana	AIML	89	Delhi
0	101	Aarav	CSE	88	Kanpur
4	105	Kabir	CSE	84	Kanpur
7	108	Meera	CSE	81	Kanpur

Task 7 — Identified missing values, counted them, demonstrated dropna, and filled missing numeric/categorical values appropriately.

Source Code

```
import pandas as pd

df = pd.read_csv("../datasets/missing_values.csv")
print("DATASET BEFORE HANDLING MISSING VALUES\n", df)

print("\nMissing values by column:\n", df.isnull().sum())

removed = df.dropna()
print("\nAfter removing rows containing missing values:\n", removed)

filled = df.copy()
filled["Marks"] = filled["Marks"].fillna(filled["Marks"].mean())
filled["Attendance"] = filled["Attendance"].fillna(filled["Attendance"].mean())
filled["Department"] = filled["Department"].fillna(filled["Department"].mode()[0])

print("\nAfter filling missing values:\n", filled)

filled.to_csv("../outputs/task07_cleaned_missing_values.csv", index=False)

print("\nWhy is handling missing data important?")
print("Missing values can distort statistics, reduce model quality, and cause errors.")
print("Appropriate removal or imputation improves data completeness and analytical reliability.")
```

Program Output

```
DATASET BEFORE HANDLING MISSING VALUES
  Name Marks Attendance Department
0  Aarav  88.0      92.0         CSE
1   Diya   NaN      85.0         CSE
2  Rohan  91.0      78.0        AIML
3 Ananya  67.0      78.0         NaN
4  Kabir   NaN      90.0         CSE
5  Ishita  72.0      86.0         ECE

Missing values by column:
Name      0
Marks     0
Attendance 2
Department 1
dtype: int64

After removing rows containing missing values:
  Name Marks Attendance Department
0  Aarav  88.0      92.0         CSE

After filling missing values:
  Name Marks Attendance Department
0  Aarav  88.0      92.0         CSE
1   Diya  79.5      85.0         CSE
2  Rohan  91.0      86.25        AIML
3 Ananya  67.0      78.0         CSE
4  Kabir  79.5      90.0         CSE
5  Ishita  72.0      86.25         ECE

Why is handling missing data important?
Missing values can distort statistics, reduce model quality, and cause errors.
Appropriate removal or imputation improves data completeness and analytical reliability.

[Stderr]
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/patches/warm_spreadsheet_runtime_on_star
rtup.py", line 26, in warm_spreadsheet_runtime_on_startup
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 785, in wa
rm_spreadsheet_runtime
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 720, in _w
arm_feature_flows
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 704, in _w
arm_collaboration_flows
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/generated/interface/models.py", line 32
317, in hydrate_crdt_from_proto
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/rpc/remote.py", line 749, in call_
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/rpc/client.py", line 150, in call_
    artifact_tool.rpc.client.RemoteError: hydrateCrdtFromProto requires an empty collaborative document.
```

Task 8 — Merged and concatenated sales data, performed GroupBy aggregations, and created a Pivot Table.

Source Code

```
import pandas as pd

jan = pd.read_csv("../datasets/sales_january.csv")
feb = pd.read_csv("../datasets/sales_february.csv")

# Merge using a common column (Product) with a small product-category lookup.
product_info = pd.DataFrame({
    "Product": ["Laptop", "Mouse", "Keyboard", "Monitor"],
    "Category": ["Computer", "Accessory", "Accessory", "Display"]
})

merged = pd.merge(pd.concat([jan, feb], ignore_index=True), product_info, on="Product", how="left")

print("MERGED DATAFRAME\n", merged)

concatenated = pd.concat([jan, feb], ignore_index=True)
print("\nCONCATENATED DATAFRAME\n", concatenated)

grouped = concatenated.groupby("Region").agg(
    Total_Sales=("Sales", "sum"),
    Average_Sales=("Sales", "mean"),
    Order_Count=("Order_ID", "count"),
    Maximum_Sale=("Sales", "max")
).reset_index()
```



```

print("\nGROUPBY SUMMARY\n", grouped)

pivot = pd.pivot_table(
    concatenated,
    values="Sales",
    index="Region",
    columns="Product",
    aggfunc="sum",
    fill_value=0
)
print("\nPIVOT TABLE\n", pivot)

merged.to_csv("../outputs/task08_merged.csv", index=False)
grouped.to_csv("../outputs/task08_groupby_summary.csv", index=False)
pivot.to_csv("../outputs/task08_pivot_table.csv")

```

Program Output

```

MERGED DATAFRAME
0  Order_ID  Product  Region  Sales  Quantity  Category
1  1001      Laptop  North  65000      1      Computer
2  1002      Mouse  South  1200      3      Accessory
3  1003  Keyboard  North  2500      2      Accessory
4  1004  Monitor  East  18000      2      Display
5  1005      Laptop  West  62000      1      Computer
6  1006      Mouse  South  1500      4      Accessory
7  1007  Keyboard  East  2700      2      Accessory
8  1008  Monitor  West  19500      2      Display
9  1009      Laptop  North  68000      1      Computer
10 1010      Mouse  East  1400      5      Accessory

CONCATENATED DATAFRAME
0  Order_ID  Product  Region  Sales  Quantity
1  1001      Laptop  North  65000      1
2  1002      Mouse  South  1200      3
3  1003  Keyboard  North  2500      2
4  1004  Monitor  East  18000      2
5  1005      Laptop  West  62000      1
6  1006      Mouse  South  1500      4
7  1007  Keyboard  East  2700      2
8  1008  Monitor  West  19500      2
9  1009      Laptop  North  68000      1
10 1010      Mouse  East  1400      5

GROUPBY SUMMARY
0  Region  Total Sales  Average Sales  Order_Count  Maximum Sale
1  East    22100      7366.666667      3          18000
2  North   135500     45166.666667      3          68000
3  South    2700      1350.000000      2           1500
4  West    81500     40750.000000      2          62000

PIVOT TABLE
Product  Keyboard  Laptop  Monitor  Mouse
Region
East      2700      0      18000     1400
North     2500     133000      0      0
South      0      62000     19500      0
West      0      62000     19500      0

[stderr]
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/patches/warm_spreadsheet_runtime_on_start
rtup.py", line 26, in warm_spreadsheet_runtime_on_startup
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 785, in wa
rm_spreadsheet_runtime
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 720, in _w
arm_feature_flows
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 704, in _w
arm_collaboration_flows
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/generated/interface/models.py", line 32
317, in hydrate_crdt_from_proto
    File "/tmp/tmp.L2TH2YScoc/artifact_tool_v2-2.8.22/artifact_tool/rpc/remote.py", line 749, in __call__

```

Task 9 — Exported processed data to CSV and verified the exported file by reading it back.

Source Code

```
import pandas as pd
from pathlib import Path

df = pd.read_csv("../outputs/task07_cleaned_missing_values.csv")
output_file = Path("../outputs/final_processed_data.csv")

df.to_csv(output_file, index=False)

print("Exported file:", output_file.resolve())
print("\nVerification - exported data:\n", pd.read_csv(output_file))
```

Program Output

```
Exported file: /mnt/data/Week_2_NumPy_Pandas_Assignment/outputs/final_processed_data.csv
Verification - exported data:
   Name  Marks  Attendance Department
0  Aarav   88.0       92.00         CSE
1   Diya   79.5       85.00         CSE
2  Rohan   91.0       86.25         AIIML
3  Ananya   67.0       78.00         CSE
4  Kabir   79.5       90.00         CSE
5  Ishita   72.0       86.25         ECE

[stderr]
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/patches/warm_spreadsheet_runtime_on_start.py", line 26, in warm_spreadsheet_runtime_on_startup
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spreadsheet_runtime
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 720, in warm_feature_flows
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/spreadsheet_warmup.py", line 704, in warm_collaboration_flows
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/generated/interface/models.py", line 32317, in hydrate_crdt_from_proto
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/remote.py", line 749, in __call__
    File "/tmp/tmp.L2TH2Y5coc/artifact_tool_v2-2.8.22/artifact_tool/rpc/client.py", line 150, in call
    artifact_tool.rpc.client.RemoteError: hydrateCrdtFromProto requires an empty collaborative document.
```

Task 10 — Completed a mini analysis of student performance including inspection, filtering, sorting, GroupBy, Pivot Table, insights and export.

Source Code

```
import numpy as np
import pandas as pd
from pathlib import Path

df = pd.read_csv("../datasets/student_performance.csv")

print("\n1. DATASET LOADED")
print(df)

print("\n2. DATA INSPECTION")
print("Shape:", df.shape)
print("Columns:", list(df.columns))
print("Missing values:\n", df.isna().sum())

# The dataset has no missing values, but this operation demonstrates safe handling.
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
df["Attendance"] = df["Attendance"].fillna(df["Attendance"].mean())

print("\n3. FILTERED STUDENTS (Marks >= 85)")
high_performers = df[df["Marks"] >= 85]
print(high_performers[["Name", "Department", "Marks", "Attendance"]])

print("\n4. SORTED BY MARKS (DESCENDING)")
print(df.sort_values("Marks", ascending=False)[["Name", "Marks"]])

print("\n5. GROUPBY DEPARTMENT")
group_summary = df.groupby("Department").agg(
    Average_Marks=("Marks", "mean"),
    Average_Attendance=("Attendance", "mean"),
    Student_Count=("Student_ID", "count")
).round(2)
print(group_summary)

print("\n6. PIVOT TABLE")
pivot = pd.pivot_table(
```

```

df,
values="Marks",
index="Department",
columns="City",
aggfunc="mean",
fill_value=0
).round(2)
print(pivot)

print("\n7. KEY INSIGHTS")
print("Overall average marks:", round(df["Marks"].mean(), 2))
print("Highest marks:", df.loc[df["Marks"].idxmax(), "Name"], "-", df["Marks"].max())
print("Lowest marks:", df.loc[df["Marks"].idxmin(), "Name"], "-", df["Marks"].min())
best_dept = df.groupby("Department")["Marks"].mean().idxmax()
print("Department with highest average marks:", best_dept)
print("Students scoring 85 or above:", len(high_performers))

out = Path("../outputs/mini_project_cleaned.csv")
df.to_csv(out, index=False)
group_summary.to_csv("../outputs/mini_project_groupby.csv")
pivot.to_csv("../outputs/mini_project_pivot.csv")
print("\nCleaned dataset exported to:", out.resolve())

```

Program Output

```

1. DATASET LOADED
Student_ID  Name  Department  Marks  Attendance  City
0 101 Aarav CSE 88 92 Kanpur
1 102 Diya CSE 76 85 Lucknow
2 103 Rohan AIML 91 96 Kanpur
3 104 Ananya AIML 67 78 Delhi
4 105 Kabir CSE 84 90 Kanpur
5 106 Ishita ECE 72 82 Lucknow
6 107 Arjun AIML 95 98 Delhi
7 108 Meera CSE 81 88 Kanpur
8 109 Vivaan ECE 69 75 Agra
9 110 Sana AIML 89 94 Delhi

2. DATA INSPECTION
Shape: (10, 6)
Columns: ['Student_ID', 'Name', 'Department', 'Marks', 'Attendance', 'City']
Missing values:
Student_ID 0
Name 0
Department 0
Marks 0
Attendance 0
City 0
dtype: int64

3. FILTERED STUDENTS (Marks >= 85)
Name  Department  Marks  Attendance
0  Aarav          CSE      88          92
2  Rohan          AIML      91          96
6  Arjun          AIML      95          98
9  Sana           AIML      89          94

4. SORTED BY MARKS (DESCENDING)
Name  Marks
6  Arjun    95
2  Rohan    91
9  Sana     89
0  Aarav    88
4  Kabir    84
7  Meera    81
1  Diya     76
5  Ishita   72
8  Vivaan   69
3  Ananya   67

5. GROUPBY DEPARTMENT
Department  Average_Marks  Average_Attendance  Student_Count
AIML        85.50        91.50                4
CSE         82.25        88.75                4
ECE         70.50        78.50                2

6. PIVOT TABLE
City  Department  Agra  Delhi  Kanpur  Lucknow
Department

```